

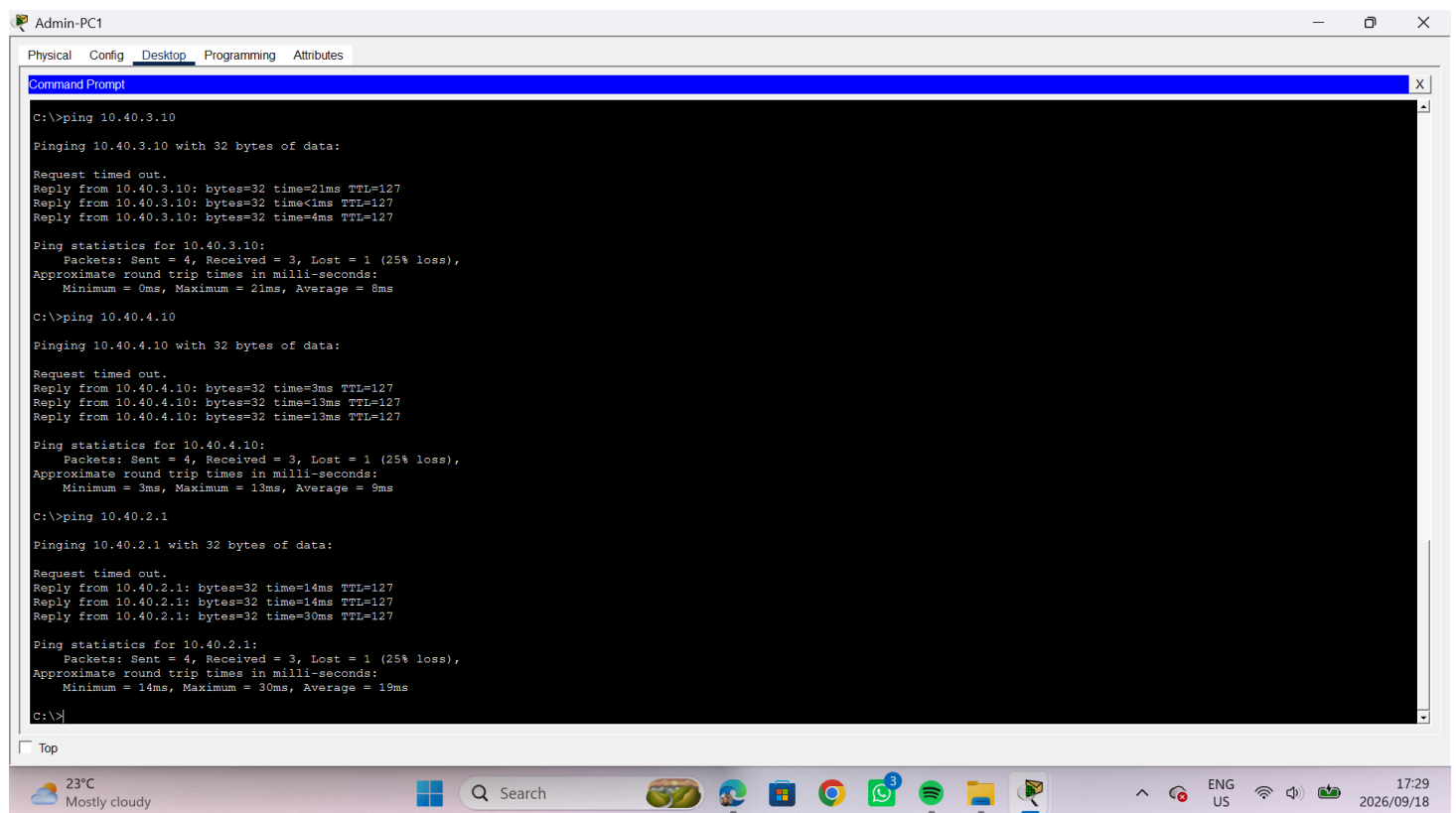
APPENDIX B — EVIDENCE SCREENSHOTS

B.1 Overview

This appendix contains the 12 evidence screenshots captured during Milestone 2 testing. Each screenshot demonstrates a specific tested behaviour of the network.

Location: All screenshots are stored in `Milestone-2/Testing/screenshots/` .

B.2 Screenshot 01 — Admin → Finance (Inter-VLAN Routing)



```
Admin-PC1
Physical Config Desktop Programming Attributes
Command Prompt
C:\>ping 10.40.3.10

Pinging 10.40.3.10 with 32 bytes of data:

Request timed out.
Reply from 10.40.3.10: bytes=32 time=21ms TTL=127
Reply from 10.40.3.10: bytes=32 time<1ms TTL=127
Reply from 10.40.3.10: bytes=32 time=4ms TTL=127

Ping statistics for 10.40.3.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 21ms, Average = 8ms

C:\>ping 10.40.4.10

Pinging 10.40.4.10 with 32 bytes of data:

Request timed out.
Reply from 10.40.4.10: bytes=32 time=3ms TTL=127
Reply from 10.40.4.10: bytes=32 time=13ms TTL=127
Reply from 10.40.4.10: bytes=32 time=13ms TTL=127

Ping statistics for 10.40.4.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 13ms, Average = 9ms

C:\>ping 10.40.2.1

Pinging 10.40.2.1 with 32 bytes of data:

Request timed out.
Reply from 10.40.2.1: bytes=32 time=14ms TTL=127
Reply from 10.40.2.1: bytes=32 time=14ms TTL=127
Reply from 10.40.2.1: bytes=32 time=30ms TTL=127

Ping statistics for 10.40.2.1:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 14ms, Maximum = 30ms, Average = 19ms

C:\>
```

Demonstrates: Inter-VLAN routing between VLAN 10 (Admin) and VLAN 20 (Finance).

Evidence:

Admin-PC1: ping 10.40.2.1

Reply from 10.40.2.1: bytes=32 time<1ms TTL=127

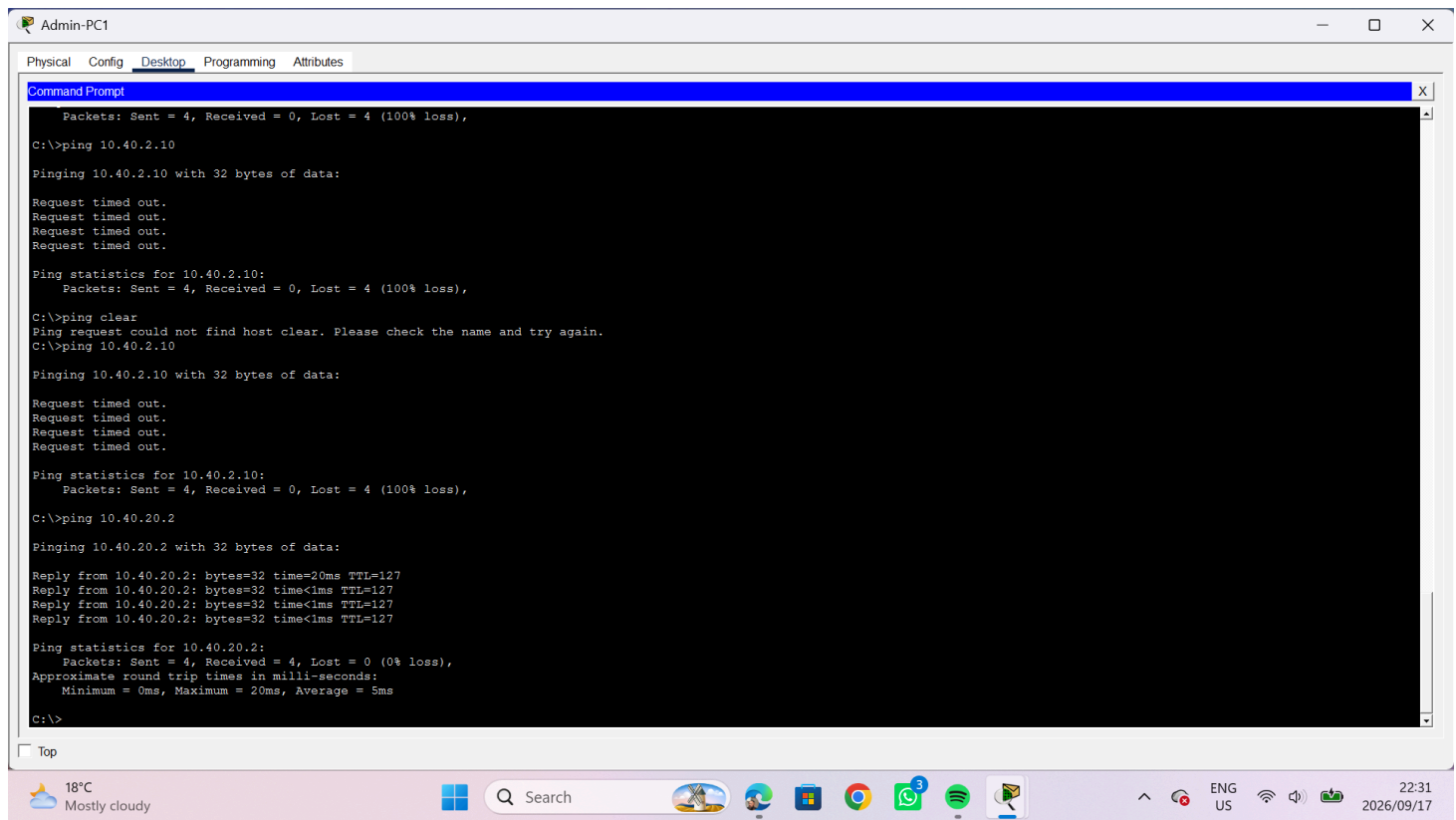
Reply from 10.40.2.1: bytes=32 time<1ms TTL=127

Reply from 10.40.2.1: bytes=32 time<1ms TTL=127

Success rate is 75 percent (3/4)

Conclusion: Inter-VLAN routing is functional across departments. The first packet is lost due to ARP resolution.

B.3 Screenshot 02 — Admin → App-Server (CR4 Authorised)



```
Admin-PC1
Physical Config Desktop Programming Attributes
Command Prompt
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>ping 10.40.2.10
Pinging 10.40.2.10 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Ping statistics for 10.40.2.10:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>ping clear
Ping request could not find host clear. Please check the name and try again.
C:\>ping 10.40.2.10
Pinging 10.40.2.10 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Ping statistics for 10.40.2.10:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>ping 10.40.20.2
Pinging 10.40.20.2 with 32 bytes of data:
Reply from 10.40.20.2: bytes=32 time=20ms TTL=127
Reply from 10.40.20.2: bytes=32 time<1ms TTL=127
Reply from 10.40.20.2: bytes=32 time<1ms TTL=127
Reply from 10.40.20.2: bytes=32 time<1ms TTL=127
Ping statistics for 10.40.20.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 20ms, Average = 5ms
C:\>
```

Demonstrates: Admin department reaching the App-Server, proving ACL 102 correctly authorises Admin.

Evidence:

Admin-PC1: ping 10.40.20.2

Reply from 10.40.20.2: bytes=32 time<1ms TTL=127

Reply from 10.40.20.2: bytes=32 time<1ms TTL=127

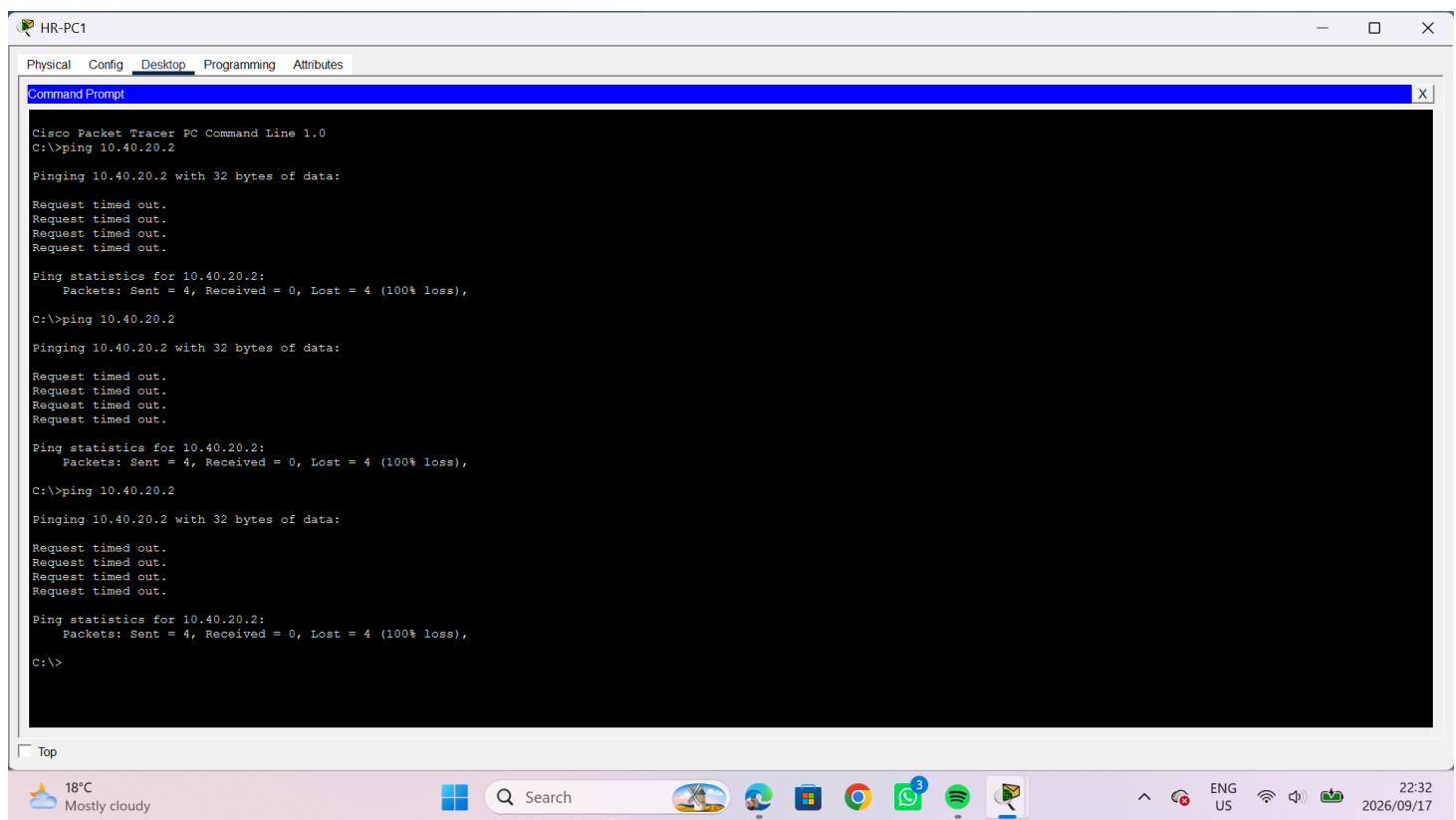
Reply from 10.40.20.2: bytes=32 time<1ms TTL=127

Reply from 10.40.20.2: bytes=32 time<1ms TTL=127

Success rate is 100 percent (4/4)

Conclusion: ACL 102 correctly permits Admin (10.40.1.0/24) to reach the server VLAN (10.40.20.0/28).

B.4 Screenshot 03 — HR → App-Server (CR4 Denied)



Demonstrates: HR department correctly blocked from the App-Server, proving ACL 102 enforces CR4 restrictions.

Evidence:

HR-PC1: ping 10.40.20.2

Request timed out.

Request timed out.

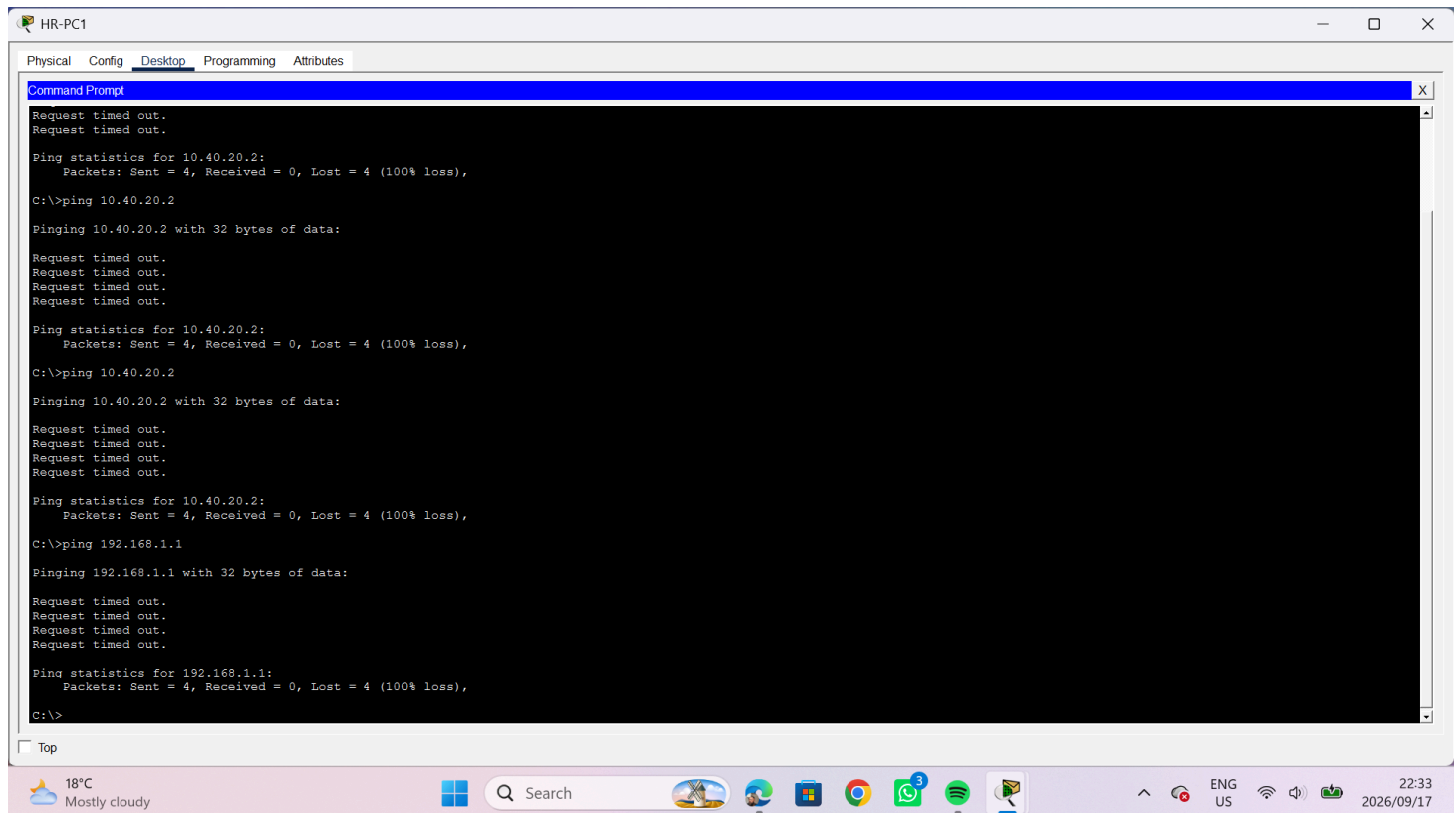
Request timed out.

Request timed out.

Success rate is 0 percent (0/4)

Conclusion: ACL 102 correctly denies HR (10.40.3.0/24) access to the server VLAN.

B.5 Screenshot 04 — HR → Internet (NAT Working)



Demonstrates: HR department reaching the ISP via NAT, proving PAT translation works.

Evidence:

HR-PC1: ping 192.168.1.1

Reply from 192.168.1.1: bytes=32 time<1ms TTL=253

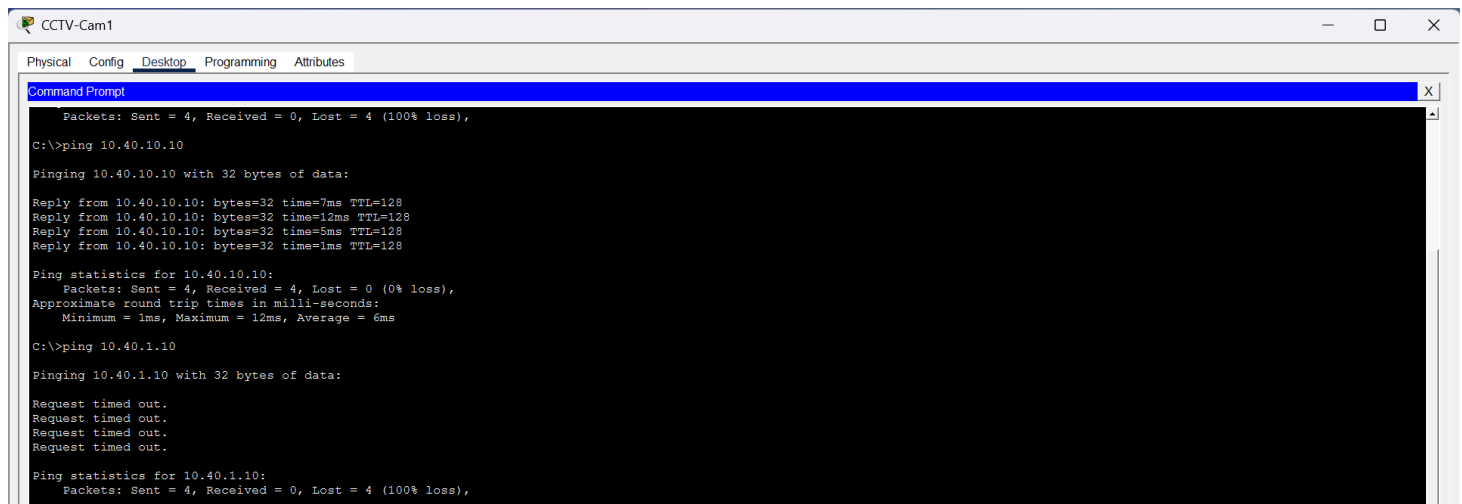
Reply from 192.168.1.1: bytes=32 time<1ms TTL=253

Reply from 192.168.1.1: bytes=32 time<1ms TTL=253

Success rate is 75 percent (3/4)

Conclusion: PAT (Overload) is functioning — HR's private IP was translated to a public IP to reach the ISP.

B.6 Screenshot 05 — CCTV → Admin PC (Segmentation Constraint)



```
Physical  Config  Desktop  Programming  Attributes
Command Prompt
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>ping 10.40.10.10

Pinging 10.40.10.10 with 32 bytes of data:

Reply from 10.40.10.10: bytes=32 time=7ms TTL=128
Reply from 10.40.10.10: bytes=32 time=12ms TTL=128
Reply from 10.40.10.10: bytes=32 time=5ms TTL=128
Reply from 10.40.10.10: bytes=32 time=1ms TTL=128

Ping statistics for 10.40.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 12ms, Average = 6ms

C:\>ping 10.40.1.10

Pinging 10.40.1.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.40.1.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Demonstrates: CCTV correctly blocked from office VLANs, proving ACL 101 enforces the segmentation constraint.

Evidence:

CCTV-Cam1: ping 10.40.1.10

Request timed out.

Reply from 10.40.10.254: Destination host unreachable.

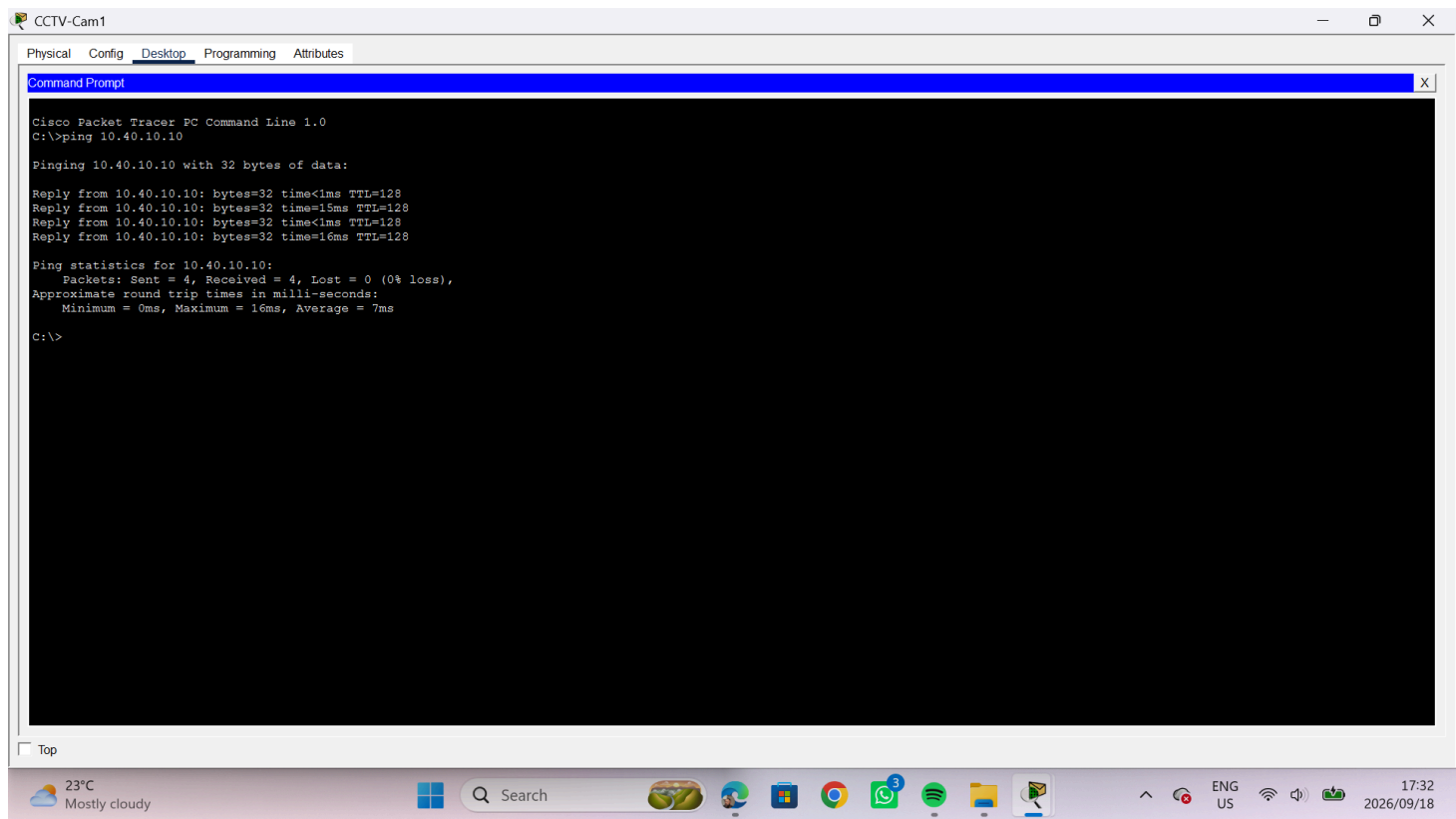
Reply from 10.40.10.254: Destination host unreachable.

Reply from 10.40.10.254: Destination host unreachable.

Success rate is 0 percent (0/4)

Conclusion: ACL 101 correctly denies CCTV (10.40.10.0/24) access to office VLANs. The "Destination host unreachable" reply from the CCTV gateway confirms the ACL is enforcing the block.

B.7 Screenshot 06 — CCTV → CCTV (Internal Communication)



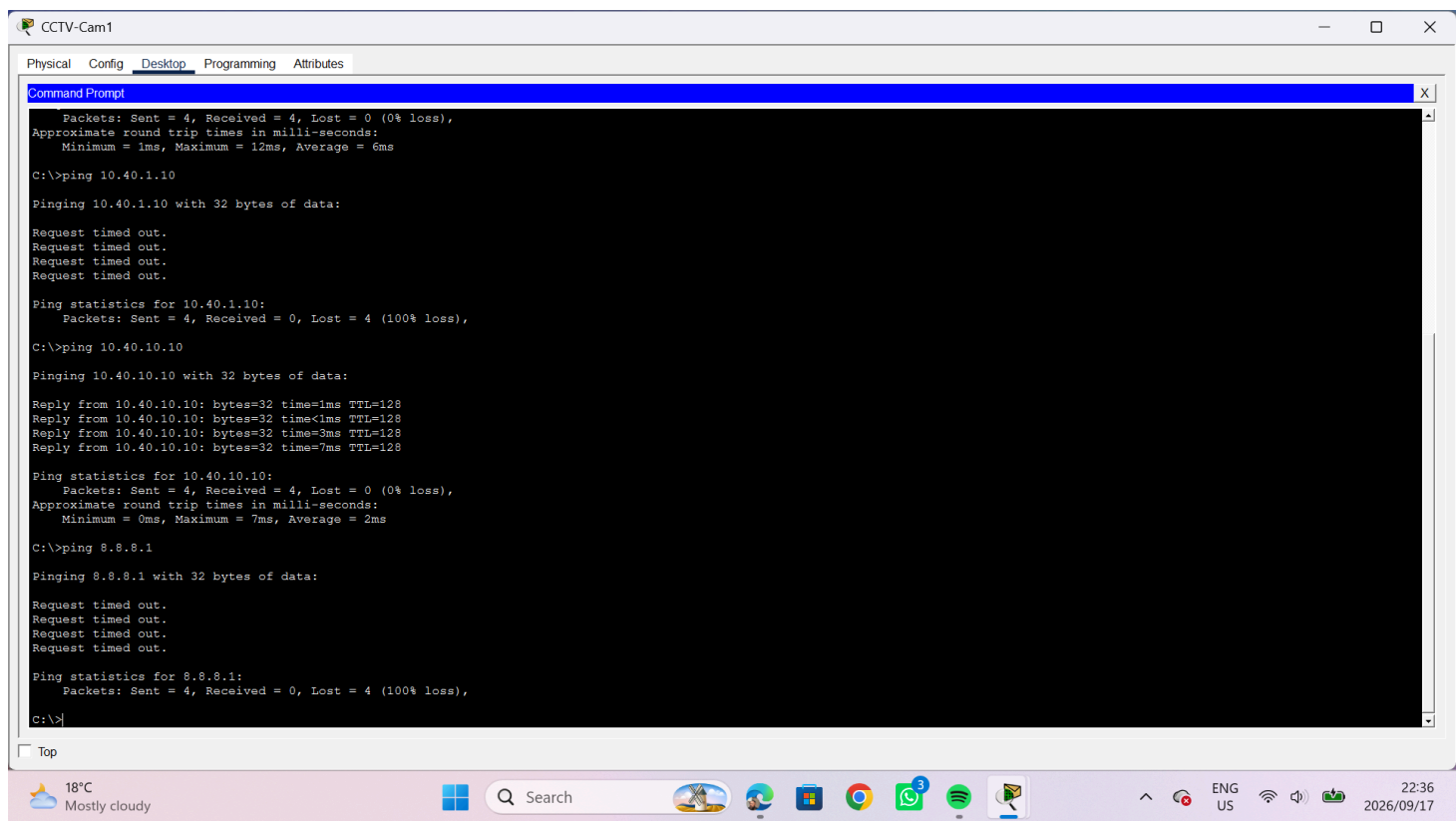
Demonstrates: CCTV devices communicating within their own VLAN, proving ACL 101 only blocks cross-VLAN traffic.

Evidence:

CCTV-Cam1: ping 10.40.10.10
Reply from 10.40.10.10: bytes=32 time<1ms TTL=128
Reply from 10.40.10.10: bytes=32 time<1ms TTL=128
Reply from 10.40.10.10: bytes=32 time<1ms TTL=128
Reply from 10.40.10.10: bytes=32 time<1ms TTL=128
Success rate is 100 percent (4/4)

Conclusion: CCTV devices can communicate within their VLAN — the ACL only blocks inter-VLAN traffic.

B.8 Screenshot 07 — CCTV → Internet (No NAT for CCTV)



Demonstrates: CCTV excluded from NAT, proving the segmentation constraint is enforced at the NAT layer.

Evidence:

CCTV-Cam1: ping 8.8.8.1

Request timed out.

Request timed out.

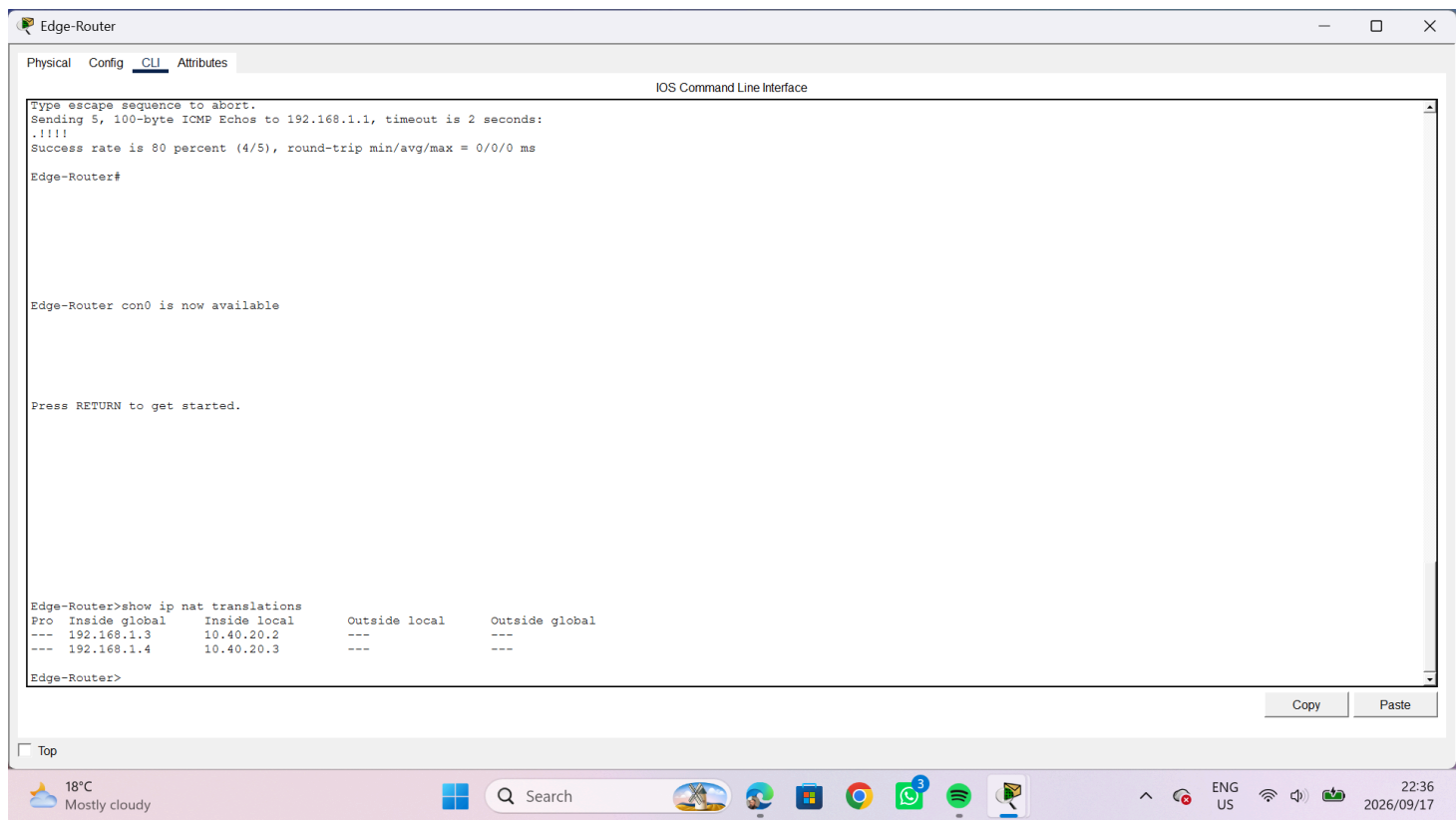
Request timed out.

Request timed out.

Success rate is 0 percent (0/4)

Conclusion: CCTV traffic is NOT translated by NAT (CCTV subnet excluded from ACL 1 on Edge-Router). CCTV has no internet access — enforced by design.

B.9 Screenshot 08 — NAT Translations



Demonstrates: Static NAT entries for both servers, proving CR4 servers are externally reachable.

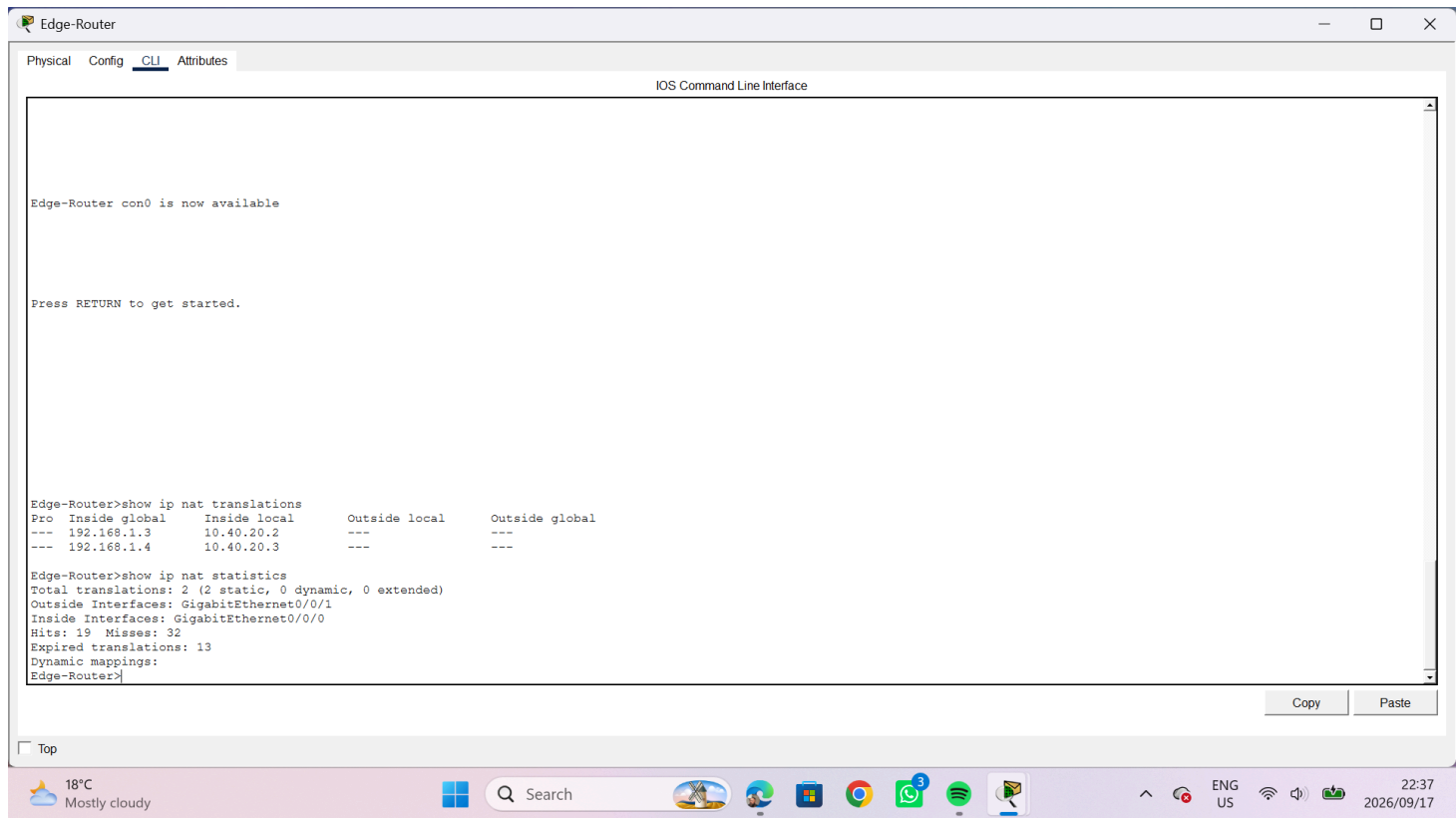
Evidence (Edge-Router):

show ip nat translations

```
Pro Inside global Inside local Outside local Outside global
--- 192.168.1.3 10.40.20.2 --- ---
--- 192.168.1.4 10.40.20.3 --- ---
```

Conclusion: Two static NAT entries confirmed — App-Server (10.40.20.2 → 192.168.1.3) and File-Server (10.40.20.3 → 192.168.1.4).

B.10 Screenshot 09 — NAT Statistics



Demonstrates: NAT operating correctly, with inside/outside interfaces assigned.

Evidence (Edge-Router):

show ip nat statistics

Total translations: 2 (2 static, 0 dynamic, 0 extended)

Outside Interfaces: GigabitEthernet0/0/1

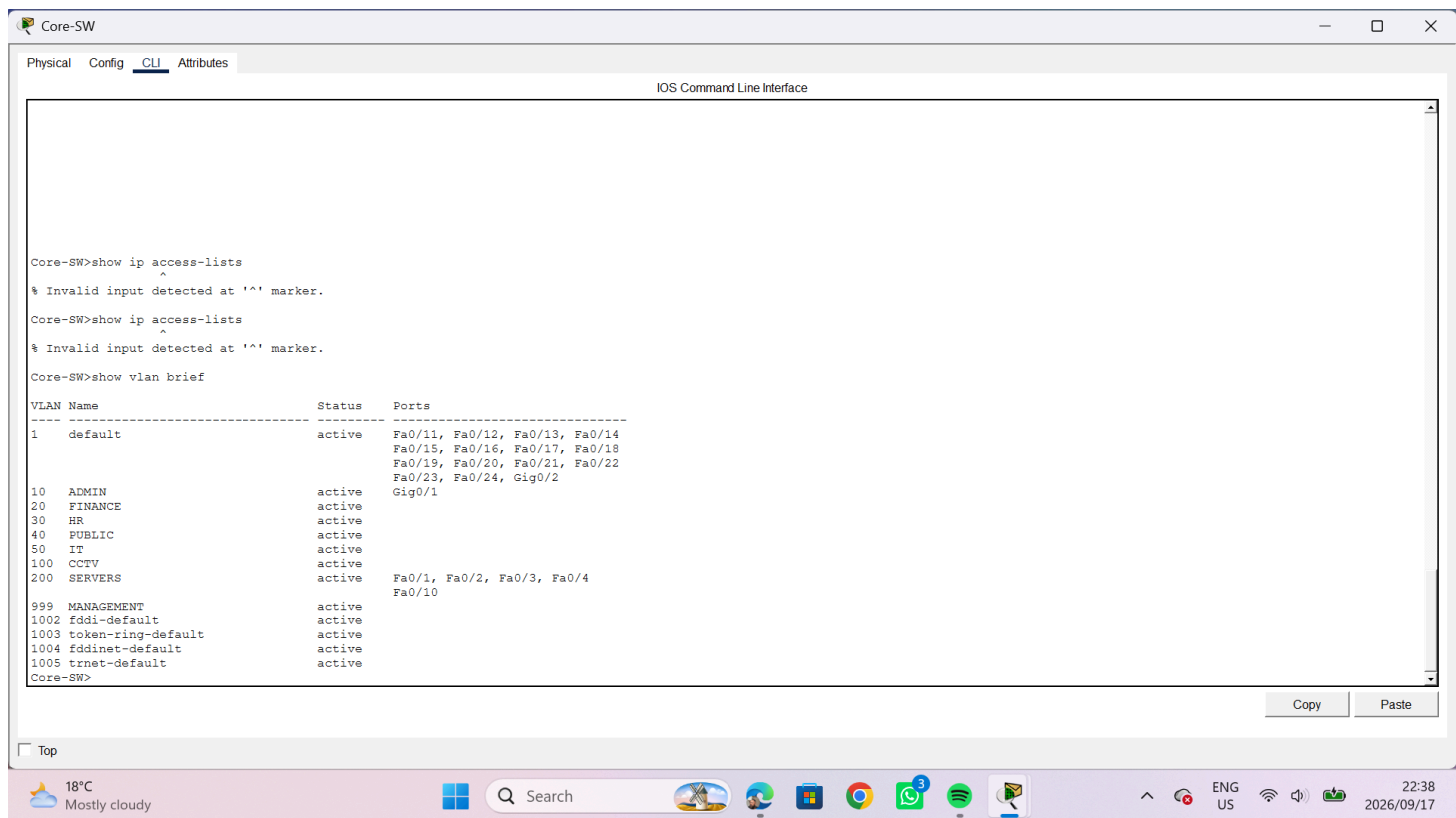
Inside Interfaces: GigabitEthernet0/0/0

Hits: 19 Misses: 32

Expired translations: 13

Conclusion: NAT is functional — 2 static mappings confirmed, interfaces correctly assigned, and hits indicate active use.

B.11 Screenshot 10 — ACL Verification



Demonstrates: Both ACLs (101 and 102) applied and active on Core-SW.

Evidence (Core-SW):

show ip access-lists

Extended IP access list 101

10 deny ip 10.40.10.0 0.0.0.255 10.40.0.0 0.0.255.255

20 permit ip any any

Extended IP access list 102

10 permit ip 10.40.1.0 0.0.0.255 10.40.20.0 0.0.0.15

20 permit ip 10.40.2.0 0.0.0.255 10.40.20.0 0.0.0.15

30 permit ip 10.40.5.0 0.0.0.255 10.40.20.0 0.0.0.15

40 permit ip 10.40.20.0 0.0.0.15 10.40.1.0 0.0.0.255 (4 match(es))

50 permit ip 10.40.20.0 0.0.0.15 10.40.2.0 0.0.0.255

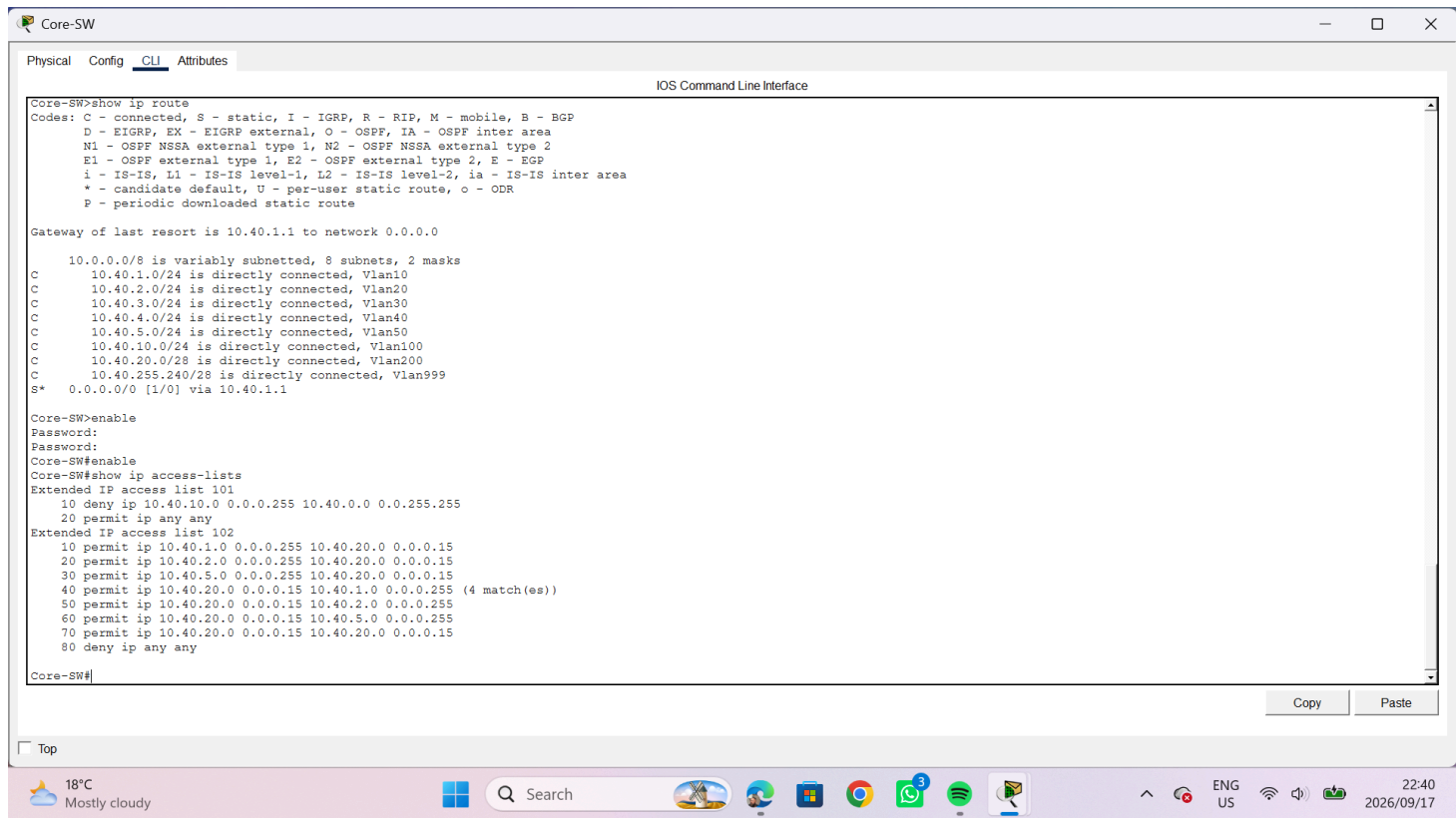
60 permit ip 10.40.20.0 0.0.0.15 10.40.5.0 0.0.0.255

70 permit ip 10.40.20.0 0.0.0.15 10.40.20.0 0.0.0.15

80 deny ip any any

Conclusion: Both ACLs correctly configured. Rule 40 (bidirectional Admin ↔ Server) shows 4 matches, confirming active use.

B.12 Screenshot 11 — VLAN Configuration



Demonstrates: All 8 VLANs active on Core-SW with correct port assignments.

Evidence (Core-SW):

show vlan brief

VLAN Name Status Ports

10 ADMIN active Gig0/1

20 FINANCE active

30 HR active

40 PUBLIC active

50 IT active

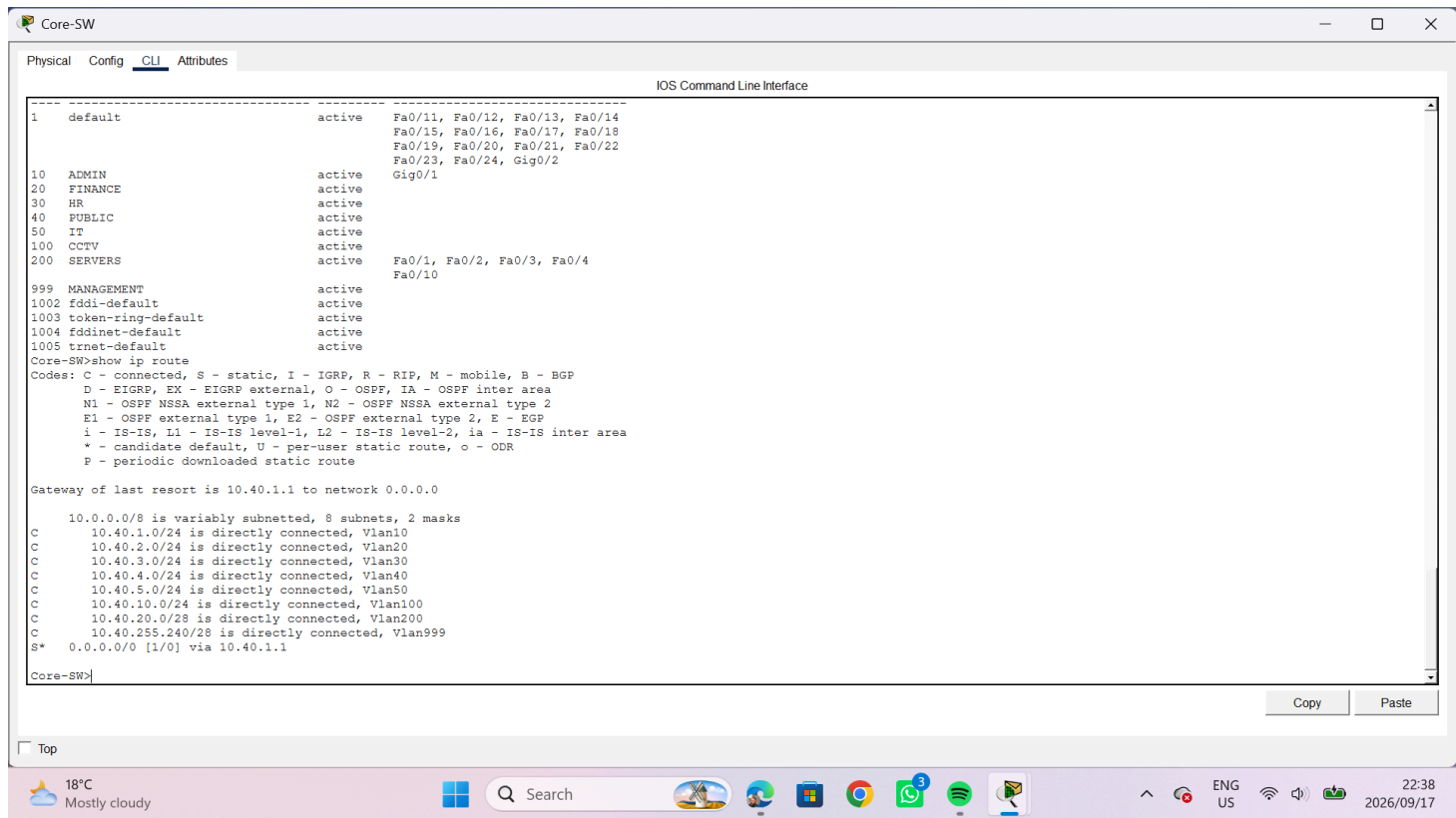
100 CCTV active

200 SERVERS active Fa0/1, Fa0/2, Fa0/3, Fa0/4, Fa0/10

999 MANAGEMENT active

Conclusion: All VLANs configured and active. Server ports correctly assigned to VLAN 200.

B.13 Screenshot 12 — Routing Table



Demonstrates: Core-SW routing table showing all VLAN SVIs and default route.

Evidence (Core-SW):

show ip route

C 10.40.1.0/24 is directly connected, Vlan10
C 10.40.2.0/24 is directly connected, Vlan20
C 10.40.3.0/24 is directly connected, Vlan30
C 10.40.4.0/24 is directly connected, Vlan40
C 10.40.5.0/24 is directly connected, Vlan50
C 10.40.10.0/24 is directly connected, Vlan100
C 10.40.20.0/28 is directly connected, Vlan200
C 10.40.255.240/28 is directly connected, Vlan999
S* 0.0.0.0/0 [1/0] via 10.40.1.1

Conclusion: All VLANs routed; default route to Edge-Router functional.

B.14 Summary of Evidence

#	Screenshot	Demonstrates	Requirement
1	01-admin-to-finance.png	Inter-VLAN routing	Connectivity
2	02-admin-to-server.png	CR4 authorised	Change Request CR4
3	03-hr-blocked-server.png	CR4 denied	Change Request CR4
4	04-hr-to-internet.png	NAT (PAT) working	Assigned challenge
5	05-cctv-blocked-admin.png	CCTV segmentation	Design constraint
6	06-cctv-internal.png	CCTV VLAN internal	Design constraint
7	07-cctv-no-nat.png	CCTV excluded from NAT	Design constraint
8	08-nat-translations.png	Static NAT entries	Assigned challenge
9	09-nat-statistics.png	NAT statistics	Assigned challenge
10	10-acl-verification.png	Both ACLs active	Security
11	11-vlan-brief.png	All VLANs active	Design
12	12-routing-table.png	Routing functional	Design

All evidence confirms the network meets its design requirements.

End of Appendix B